clk

rst_n

u_hazard

hazard_detection

ex_mem_read
ex_rd_addr[4:0] stall
id_rs1_addr[4:0]
id_rs2_addr[4:0]

u_imem
instruction_memory

addr[31:0] instruction[31:0]

pc_plus4_in1_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

pc_in1_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

instruction_in1_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

pc_plus4_in0_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

instruction_in0_i RTL_MUX S

V=X"00000013", S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

pc_in0_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

u_if_id

if_id_reg

clk

rst_n

instruction_in[31:0] instruction_out[31:0] pc_in[31:0] pc_out[31:0] pc_plus4_in[31:0] pc_plus4_out[31:0]

u_ctrl

control_unit

alu_src branch

jump mem_read mem_to_reg mem_write

reg_write

alu_op[1:0] opcode[6:0]

reg_write_in2_i RTL_OR

I0

I1 O

branch_in0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

jump_in0_i RTL_MUX

S=1'b1 I0

S=default I1 O S mem_read_in0_i RTL_MUX

S=1'b1 I0

S=default I1 O S mem_write_in0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

reg_write_in0_i RTL_MUX

S=1'b1 I0

S=default I1 O S

u_rf

register_file

clk

reg_write
rst_n

rd_addr[4:0] rd_data[31:0]

rs1_addr[4:0]

rs1_data[31:0]

rs2_addr[4:0]

rs2_data[31:0]

u_immgen
immediate_generator

instruction[31:0] imm_out[31:0]

u_id_ex

id_ex_reg

alu_src_in

branch_in alu_src_out clk branch_out

jump_in jump_out mem_read_in mem_read_out mem_to_reg_in mem_to_reg_out mem_write_in mem_write_out

reg_write_in reg_write_out rst_n

alu_op_in[1:0] alu_op_out[1:0]

funct3_in[2:0] funct3_out[2:0]
funct7_in[6:0] funct7_out[6:0]
imm_out_in[31:0] imm_out_out[31:0]

opcode_in[6:0] opcode_out[6:0] pc_in[31:0] pc_out[31:0] pc_plus4_in[31:0] pc_plus4_out[31:0] rd_addr_in[4:0] rd_addr_out[4:0]

rs1_addr_in[4:0] rs1_addr_out[4:0] rs1_data_in[31:0] rs1_data_out[31:0] rs2_addr_in[4:0] rs2_addr_out[4:0] rs2_data_in[31:0] rs2_data_out[31:0] u_fwd

forwarding_unit mem_reg_write wb_reg_write

ex_rs1_addr[4:0] ex_rs2_addr[4:0]
forward_a[1:0]

forward_b[1:0] mem_rd_addr[4:0] wb_rd_addr[4:0]

ex_rs2_forwarded0_i RTL_MUX

S=2'b01 I0[31:0] S=default I1[31:0] O[31:0] S[1:0]

ex_rs2_forwarded_i RTL_MUX

S=2'b10 I0[31:0] S=default I1[31:0] O[31:0] S[1:0]

u_ex_mem

ex_mem_reg

clk

jump_in jump_out mem_read_in mem_read_out mem_to_reg_in mem_to_reg_out mem_write_in mem_write_out
reg_write_in reg_write_out rst_n

alu_result_in[31:0] alu_result_out[31:0]

pc_plus4_in[31:0] pc_plus4_out[31:0] rd_addr_in[4:0] rd_addr_out[4:0]
rs2_data_in[31:0] rs2_data_out[31:0]

ex_rs1_forwarded0_i RTL_MUX

S=2'b01 I0[31:0] S=default I1[31:0] O[31:0] S[1:0] =

is_auipc_i RTL_EQO

I0[6:0] V=B"0010111" I1[6:0]

ex_rs1_forwarded_i RTL_MUX

S=2'b10 I0[31:0] S=default I1[31:0] O[31:0] S[1:0]

u_aluctrl

alu_control

alu_ctrl[3:0] alu_op[1:0]

funct3[2:0]

funct7[6:0] opcode[6:0]

ex_alu_operand_b_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

ex_alu_operand_a_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

u_alu

alu

zero_flag

alu_op[3:0] alu_result[31:0] operand_a[31:0] operand_b[31:0]

ex_branch_base_i RTL_MUX

S=2'b00 I0

S=default I1 O

S[1:0]

branch_taken0_i RTL_XOR

I0

I1 O =

is_jalr_i RTL_EQO
I0[6:0] V=B"1100111" I1[6:0] +

pc_target_i RTL_ADD

I0[31:0]

I1[31:0] O[31:0]

branch_taken_i RTL_AND

I0

I1 O

pc_redirect_target0_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

flush_i RTL_OR

I0

I1 O

pc_redirect_target_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

if_pc_next0_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

wb_reg_write_data0_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

u_dmem

data_memory

clk mem_read mem_write

addr[31:0] read_data[31:0] write_data[31:0]

if_pc_next_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

wb_reg_write_data_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

u_mem_wb

mem_wb_reg

clk

jump_in jump_out mem_to_reg_in mem_to_reg_out

reg_write_in reg_write_out rst_n

alu_result_in[31:0] alu_result_out[31:0] mem_read_data_in[31:0] mem_read_data_out[31:0] pc_plus4_in[31:0] pc_plus4_out[31:0] rd_addr_in[4:0] rd_addr_out[4:0]

u_pc

program_counter

clk
rst_n

pc_next[31:0] pc_out[31:0] pc_plus4[31:0]

debug_wb_data[31:0]

debug_wb_rd[4:0] debug_wb_write

debug_pc[31:0]